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Network Tomography: Estimating Source-Destination Traffic Intensities from Link Data

Authors	Y. Vardi

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Abstract

The problem of estimating the node-to-node traffic intensity from repeated measurements of traffic on the links of a network is formulated and discussed under Poisson assumptions and two types of traffic-routing regimens: deterministic (a fixed known path between each directed pair of nodes) and Markovian (a random path between each directed pair of nodes, determined according to a known Markov chain fixed for that pair). Maximum likelihood estimation and related approximations are discussed, and computational difficulties are pointed out. A detailed methodology is presented for estimates based on the method of moments. The estimates are derived algorithmically, taking advantage of the fact that the first and second moment equations give rise to a linear inverse problem with positivity restrictions that can be approached by an EM algorithm, resulting in a particularly simple solution to a hard problem. A small simulation study is carried out.

Real-World Impact & Application

AT&T Nationwide Traffic Matrix Estimation: Managing continental telecom backbones requires knowing the volume of traffic flowing between every origin and destination city. Yehuda Vardi formulated this problem as 'network tomography', using statistical inverse problem methods (expectation-maximization) to deduce origin-destination traffic demands solely from simple link byte counters. This remains a cornerstone of ISP traffic engineering.